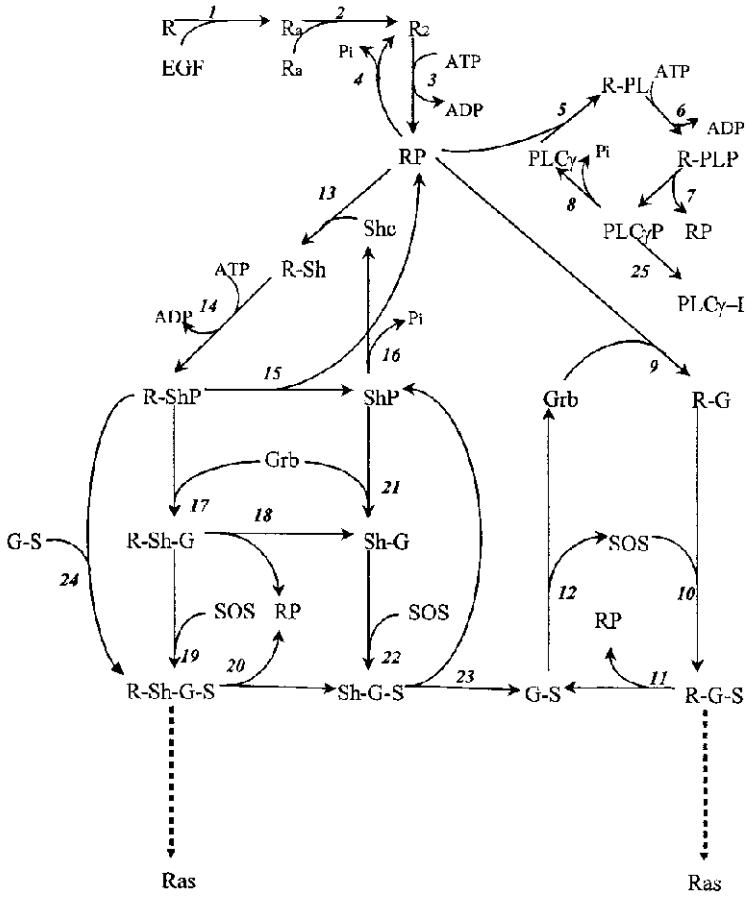
Reaction Network Models

Reaction Network Scheme ⟺ Mathematical Formulation



Rate Equations

$k_1 \cdot [R] \cdot [EGF] - k_{-1} \cdot [R_a]$
$k_2 \cdot [R_a] \cdot [R_a] - k_{-2} \cdot [R_2]$
$k_3 \cdot [R_2] - k_{-3} \cdot [RP]$
$V_4 \cdot [RP]/(K_4 + [RP])$
$k_5 \cdot [RP] \cdot [PLC\gamma] - k_{-5} \cdot [R\text{-}PL]$
$k_6 \cdot [R\text{-}PL] - k_{-6} \cdot [R\text{-}PLP]$
$k_7 \cdot [R\text{-}PLP] - k_{-7} \cdot [RP] \cdot [PLC\gamma P]$
$V_8 \cdot [PLC\gamma P]/(K_8 + [PLC\gamma P])$
$k_9 \cdot [RP] \cdot [Grb] - k_{-9} \cdot [R\text{-}G]$
$k_{10} \cdot [R\text{-}G] \cdot [SOS] - k_{-10} \cdot [R\text{-}G\text{-}S]$
$k_{11} \cdot [R\text{-}G\text{-}S] - k_{-11} \cdot [RP] \cdot [G\text{-}S]$
$k_{12} \cdot [G\text{-}S] - k_{-12} \cdot [Grb] \cdot [SOS]$
$k_{13} \cdot [RP] \cdot [Shc] - k_{-13} \cdot [R\text{-}Sh]$
$k_{14} \cdot [R\text{-}Sh] - k_{-14} \cdot [R\text{-}ShP]$
$k_{15} \cdot [R\text{-}ShP] - k_{-15} \cdot [ShP] \cdot [RP]$
$V_{16} \cdot [ShP]/(K_{16} + [ShP])$
$k_{17} \cdot [R\text{-}ShP] \cdot [Grb] - k_{-17} \cdot [R\text{-}Sh\text{-}G]$
$k_{18} \cdot [R\text{-}Sh\text{-}G] - k_{-18} [RP] \cdot [Sh\text{-}G]$
$k_{19} \cdot [R\text{-}Sh\text{-}G] \cdot [SOS] - k_{-19} \cdot [R\text{-}Sh\text{-}GS]$
$k_{20} \cdot [R\text{-}Sh\text{-}G\text{-}S] - k_{-20} \cdot [Sh\text{-}G\text{-}S] \cdot [RP]$
$k_{21} \cdot [ShP] \cdot [Grb] - k_{-21} \cdot [Sh\text{-}G]$
$k_{22} \cdot [Sh\text{-}G] \cdot [SOS] - k_{-22} \cdot [Sh\text{-}G\text{-}S]$
$k_{23} \cdot [Sh\text{-}G\text{-}S] - k_{-23} \cdot [ShP] \cdot [G\text{-}S]$
$k_{24} \cdot [R\text{-}ShP] \cdot [G\text{-}S] - k_{-24} \cdot [R\text{-}Sh\text{-}G\text{-}S]$
$k_{25} \cdot [PLC\gamma P] - k_{-25} \cdot [PLC\gamma P\text{-}I]$

Differential Equations

$d[EGF]/dt = -v_1$
$d[R]/dt = -v_1$
$d[R_a]/dt = v_1 - 2v_2$
$d[R_2]/dt = v_2 + v_4 - v_3$
$dRP/dt = v_3 + v_7 + v_{11} + v_{15} + v_{18} + v_{20} - v_4 - v_5 - v_9 - v_{13}$
$d[R\text{-}PL]/dt = v_5 - v_6$
$d[R\text{-}PLP]/dt = v_6 - v_7$
$d[R\text{-}G]/dt = v_9 - v_{10}$
$d[R\text{-}G\text{-}S]/dt = v_{10} - v_{11}$
$d[R\text{-}Sh]/dt = v_{13} - v_{14}$
$d[R\text{-}ShP]/dt = v_{14} - v_{24} - v_{15} - v_{17}$
$d[R\text{-}Sh\text{-}G]/dt = v_{17} - v_{18} - v_{19}$
$d[R\text{-}Sh\text{-}G\text{-}S]/dt = v_{19} - v_{20} + v_{24}$
$d[G\text{-}S]/dt = v_{11} + v_{23} - v_{12} - v_{24}$
$d[ShP]/dt = v_{15} + v_{23} - v_{21} - v_{16}$
$d[Sh\text{-}G]/dt = v_{18} + v_{21} - v_{22}$
$d[PLC\gamma]/dt = v_8 - v_5$
$d[PLC\gamma P]/dt = v_7 - v_8 - v_{25}$
$d[PLC\gamma P\text{-}I]/dt = v_{25}$
$d[Grb]/dt = v_{12} - v_9 - v_{17} - v_{21}$
$d[Shc]/dt = v_{16} - v_{13}$
$d[SOS]/dt = v_{12} - v_{10} - v_{19} - v_{22}$

22 species / 25 reactions

Kholodenko et al., *J. Biol. Chem.* (1999)